

U-Net and Watershed Based Deep Learning Approach for Panoptic Segmentation of Colorectal Polyps in Colonoscopy

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Abstract—Colon cancer has emerged as one of the deadliest of its kind contributing to a high mortality rate. This starts with the formation of a pouch-like structure called polyp and early detection of this polyp can help in reducing the mortality rate significantly. Detecting polyps manually is a hectic and erroneous process as some polyps are small and look like their surroundings. Deep learning-based automatic detection of polyps can help this cause and make the process time-efficient using various segmentation models. Traditional segmentation lacks the capability of detecting multiple polyps in a single area as this shows gross area affected only. To overcome these lacking we proposed a deep learning-based approach combining a modified U-Net model and Watershed algorithm to segment the colonoscopy images. This produces mask files in a colored format where the boundaries among multiple polyps are well-defined and indicated by different colors. This helps in diagnosing a patient well and reduces the miss rate for polyps. Our proposed work has been trained on the Kvasir-SEG dataset and implemented on Kvasir V2 polyp class data for segmentation. Our combination of U-Net and Watershed algorithm returned an accuracy of 99.24% and 94.83% for training and validation, respectively. Our primary target is to detect, segment, and determine the size and number of polyps with precision to automate this whole process which is backed by the accuracy and other metrics obtained from the experiment.

Index Terms—Panoptic segmentation, deep learning, Watershed algorithm, gastrointestinal tract, Kvasir-SEG dataset, Kvasir V2 dataset.

I. INTRODUCTION

Colorectal Cancer (CRC) is the third most contributing disease to the mortality rate worldwide [1]. This case is even worse in the case of a third-world country like Bangladesh. This is considered the number one in terms of death rate due to cancer in Bangladesh as this country is overpopulated with respect to the number of physicians [2]. However, this mortality rate can be reduced if early detection of cancer can be done. Polyp is the early stage of this type of cancer and needs to be identified with precise information to diagnose a patient well. Without proper detection, this turns into malignant cancer and there is very little probability of getting cured from there as with the advancement of stages of cancer it gets spread in healthy cells as shown in Fig. 1 [3].

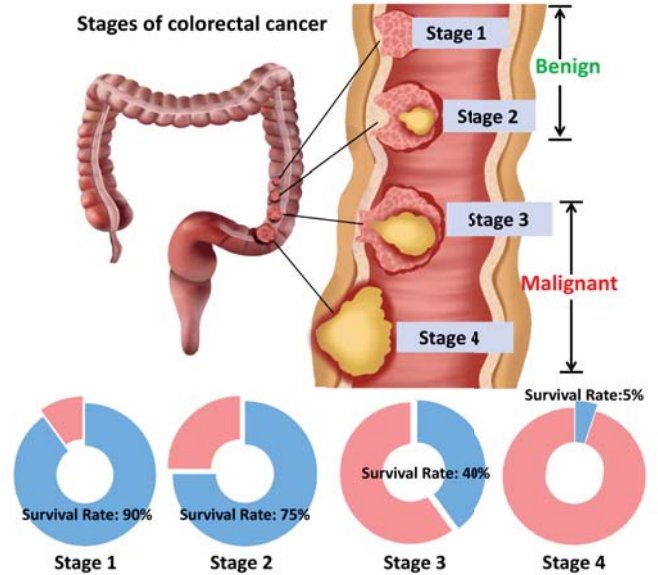


Fig. 1. Stages of CRC [3].

A colonoscopy is conducted to analyze abnormalities in the human small intestine, large intestine, colon, and rectum. A camera and a tube are inserted in these areas to record video as well as to take still images which are processed later to analyze further abnormalities [4]. But doing this analysis manually is as hard as it can get. Time is another constraint in the manual process as the number of patients is high compared to the number of physicians available for this task. Also, these processes require experts in colonoscopy which is also not in abundance in third-world countries. To overcome these shortcomings researchers are working heart and soul to make this whole process automatic and less human-dependent.

Semantic segmentation is a way to make this detection of polyps automatic and more reliable. In semantic segmentation the affected area is denoted by a pixel value of 255 and the rest of the area is denoted by a pixel value of 0. Initially, traditional machine learning algorithms like Random Forest and Decision Tree were used for this semantic segmentation. However,

these did not prove to be efficient as the surroundings of the colorectal area are such that it is very difficult to differentiate polyps. Gradually deep learning-based models also started to come into the picture with time. U-Net is one of the most anticipated CNN models with high accuracy and efficiency in the case of medical imaging. This plays a significant role in any type of medical image segmentation. ResNet, Segnet, and R-CNN are some other models that are widely used in this area of semantic segmentation.

Semantic segmentation reduces the manual intervention and the hardship associated with polyp detection. However, this still is not sufficient enough to diagnose a patient well because semantic segmentation only indicates or separates the total area where polyps are located. Sometimes it is hard to detect small polyps just beside a big one as the whole area is merged in semantic segmentation. Also, the size of each polyp cannot be understood which hinders proper treatment in case multiple polyps are available in the same area. Here comes the role of instance segmentation where each object of the same class is defined using different pixel values and putting a boundary among the overlapping or merging objects. The Watershed algorithm is one of the most used algorithms for instance segmentation. A combination of both semantic and instance segmentation is termed panoptic segmentation. First, the images are defined according to their class and later on, the objects of the same class are also segmented using instance segmentation.

In this work, we propose an improved panoptic segmentation model based on the U-Net and Watershed algorithm, which can segment the polyps in colposcopy images as well as it can define how many polyps are there along with their size. We used the Kvasir-SEG dataset [5] to train our model and later tested it on the polyp class data of the Kvasir V2 dataset [6]. This data does not contain the ground truth for the polyp data to date and researchers are trying to accurately segment them. Our work is a part of this process with more clarity provided by the inclusion of instance segmentation.

II. LITERATURE REVIEW

Early detection of CRC has been one of the main tasks for researchers in the field of segmentation. Various groups have proposed techniques to localize polyps in the human colorectal area. Many new approaches have been tested in CNN models and deep learning methods.

Polyp segmentation has seen a major jump towards accuracy with the introduction of U-Net by Ronneberger et al. [7]. This raised the standard of medical image segmentation and paved the way for researchers to experiment more with this U-Net and build customized models. Fan et al. [8] proposed a reverse attention module-based model named PraNet which performed with more accuracy than the existing models. They did the mining of boundary information and a combination of global features to achieve this accuracy.

Jha et al. [9] proposed ResUNet++ achieving an Intersection over Union (IoU) value of 79.27% and 79.62% on Kvasir-SEG and CVC-612 datasets, respectively. They blended the ResNet

and U-Net algorithms and further changed the layers of U-Net to perform this experiment. Guo and Matuszewski [10] developed and tested two structures for polyp segmentation. One of them contained deep residual network and dilation kernels, and the other contained squeeze and extraction modules along with augmented U-Net. Lou et al. [11] developed Context Axial Reserve Attention Network (CaraNet) which focused on small objects and achieved remarkable accuracy with CVC-DB dataset. They achieved a dice coefficient of 0.936 and an IoU score of 0.887 for this dataset.

Wang et al. [12] proposed an inverse attention mechanism blended with multiscale fusion networks to develop a new model for boundary detection of polyps in colonoscopy images. They experimented with their model on several datasets like Kvasir, CVC-ClinicDB, CVC-300, ETIS LaribPolypDB, etc. Residual and atrous convolution in parallel unit-based U-Net was used by Lee and Yoo [13] which gave a better result in terms of mean IoU and dice score on the same datasets used by Wang et al. [12]. Shen et al. [14] combined the HarDNet module and multi-coding attention mechanism to optimize the two stages of encoding and decoding which helped in increasing the accuracy of the model in polyp segmentation and scored 89.64%, 94.91%, and 99.65% accuracy on Kvasir-SEG, ISIC2018, and COVID-19CT datasets, respectively.

Uncertainty exploration was done by Guo et al. [15]. They divided the pixels of an image into three categories namely background, foreground, and uncertain area. Uncertain area boundaries are more highlighted and features are extracted so that it can perform better in polyps segmentation where foreground and background pixels are kind of similar. Sharma et al. [16] developed a U-Net-based model that can detect polyps in gastrointestinal tract images. This model outperformed several pre-trained models, like Inception V3, VGG19, ResNet50, and DenseNet in terms of accuracy and loss measurements.

III. METHODOLOGY

In our work, we proposed a hybrid model where a simplified U-Net has been used followed by an optimized Watershed algorithm to segment polyps from the colonoscopy images. An overview of our proposed model is shown in Fig. 2. This model consists of four major segments which are described below.

A. Preprocessing of Data

Data preprocessing is one of the major activities when it comes to image segmentation because segmentation can mess things up if images are not preprocessed as per the requirement of the model. We took the images from the dataset as input and preprocessed them before feeding them to our model. We converted them from the default BGR format to RGB format as we used Open CV as our image reading library. Next, we resized them to our desired 256×256 pixel size. Later the images were normalized and appended in an array format required for feeding to our model. This dataset was then split into training and validation sections for training and subsequent validation of our model.

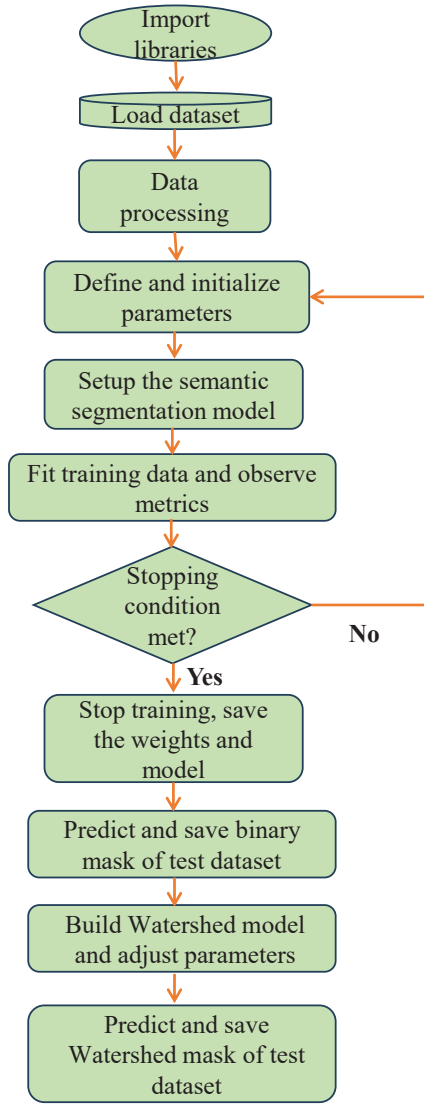


Fig. 2. Framework for the proposed panoptic segmentation.

B. Building and Training the Model

We have chosen the U-Net model as it is a proven solution for segmentation tasks, especially for medical image data. This outperforms other models in terms of accuracy and IoU metrics. U-Net models built by researchers over time have taken a complex form by including various CNN layers. Our target is to simplify this model so that we can save significant computational power consumption by the model. It also gives us room to simplify the hardware requirements when we want to implement this model in practical life. Our simplified model structure is shown in Fig. 3 describing the size of data in each layer along with the activation function used for respective layers. Several encoder and decoder blocks were used along with bridging or concatenation layers to form the first part of the model.

We have used convolution2D layers along with batch normalization, pooling, dropout, and activation layers as part of

the U-Net model. Total trainable and non-trainable parameters for this whole model were 31,095,873 and 6,016 respectively. We did not use any dense layer which itself reduces the complexity of any model. In this part of our model, we developed only the portion that does semantic segmentation of our input dataset. This produces the binary mask of our image data. We have used Adam optimizer and Binary cross-entropy as our loss function for this U-Net portion. A learning rate reducer is also used to fine-tune the model weights, depending on the validation accuracy during the learning process.

C. Testing and Generating Binary Mask

After finishing the training part, we tested our model on polyp images from the Kvasir V2 dataset to see if our model segmented the polyps properly or not. According to the results, we adjusted our model hyperparameters and trained the model again till we got a satisfactory result. After finalizing our model we generated binary masks for all the polyp images of Kvasir V2 and dumped them in a new folder which is also a milestone as there is no proven ground truth for this data to date. These binary masks saved in a different folder were used later for instance segmentation which is the second part of our work.

D. Applying Watershed Algorithm

The saved binary mask files were imported again to the final part of our work for instance segmentation. This algorithm also required preprocessed images as its input. We again preprocessed the binary masks and then did some morphological operations as part of our Watershed algorithm as per our requirement. We kept the operations as simple as they could be and developed a simplified algorithm for the Watershed. Later we put the mask files through the Watershed function and produced Watershed masks containing multiple pixel values

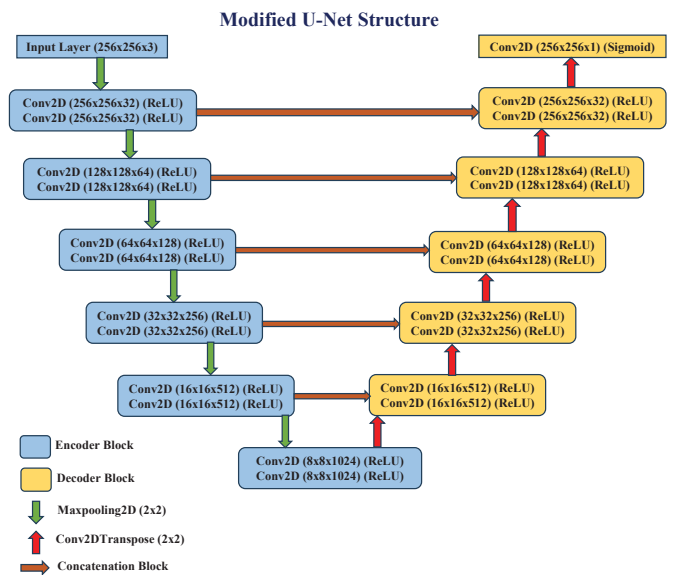


Fig. 3. Modified U-Net diagram with kernel size.

for multiple polyps including their boundary line. This way we can differentiate the number of polyps along with their area and shape in a particular area. As we implemented both the semantic and instance segmentation in our model thus we call this a panoptic segmentation model.

IV. RESULTS AND DISCUSSION

First, we trained our model on the Kvasir-SEG dataset and after training, we tested the model on the Kvasir V2 polyp image data to determine its validity. We used metrics such as accuracy, loss, and IoU to compare our model with other state-of-the-art models.

A. Dataset Details

We have trained our model on the Kvasir-SEG dataset which contains 1000 images of polyps in colorectal area [5]. These images were taken using colonoscopes at Vestre Viken Health Trust in Norway from hundreds of patients. The size of these images is also different because the camera and orientation used were different from time to time during the diagnosis of patients. After preprocessing, some sample images with their corresponding masks are shown in Fig. 4. This dataset also contains the corresponding binary mask of the same size for each image. 90% of this data was used for training our model and the rest 10% was used to do the validation.

We also used the Kvasir V2 dataset [6] to test our model, which consists of 1,000 images in each of the 8 classes of endoscopic images. These images are subdivided into three major groups namely anatomical landmarks, pathological findings, and stages of polyp removal. However, we have used only the polyp class data as our prime target is to segment

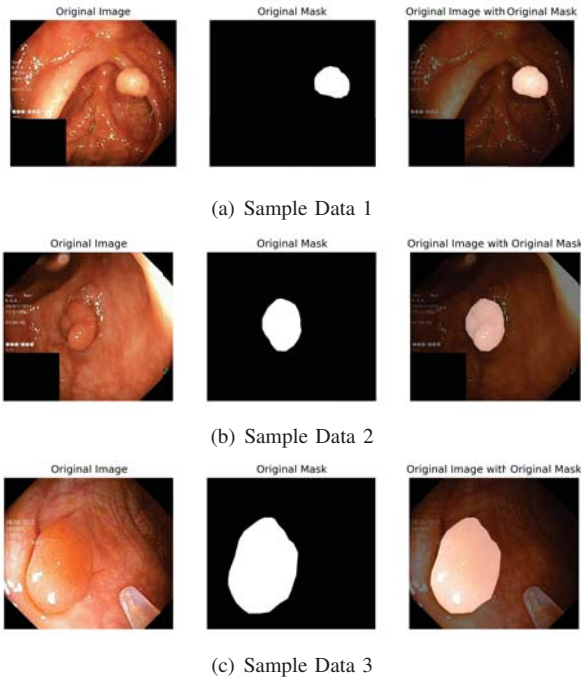


Fig. 4. Dataset with ground truths.

the polyp images. Additionally, this dataset did not have any reference ground truth or binary masks. We produced binary as well as Watershed masks for the whole dataset and our model performed pretty well.

B. Hardware Setup

Python was the primary programming language used to build our modified model using Anaconda Navigator. We used the TensorFlow (Version: 2.10.0) environment which we installed previously along with Keras (Version: 2.10.0). The compiler we used for this whole proceeding is Jupyter Notebook which is a part of Anaconda Distribution. A central processing unit with an 8th generation i5 processor was used to install the above software and run them simultaneously. The processor was a product of Intel and the model was 8265U. Intel provided this processor with a base speed of 1.60 GHz, which can reach a maximum of 1.80 GHz. Windows 10 Pro from Microsoft Corporation was used with an OS build of number 19045.5011. Our entire system was supported by a Dell laptop with 8 GB of RAM, a 256 GB SSD, and a 1 TB HDD, which was sufficient for our computational needs.

C. Accuracy and Loss Curves

The accuracy and loss curves are one of the major indications of how accurately a model can perform and Fig. 5(a) and 5(b), respectively show us the accuracy and loss profile of our proposed model. These curves express the stability of both accuracy and loss after a very minimum number of epochs. After applying the modified U-Net algorithm the binarized

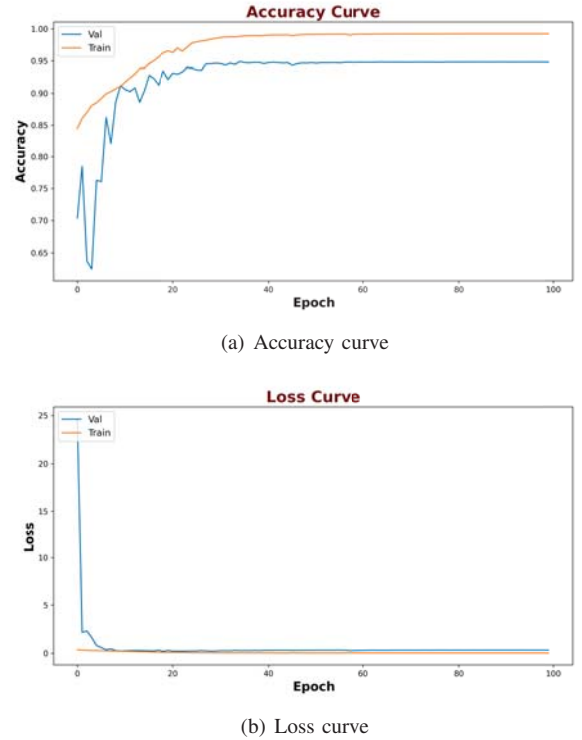


Fig. 5. Accuracy and loss curves for modified semantic segmentation model.

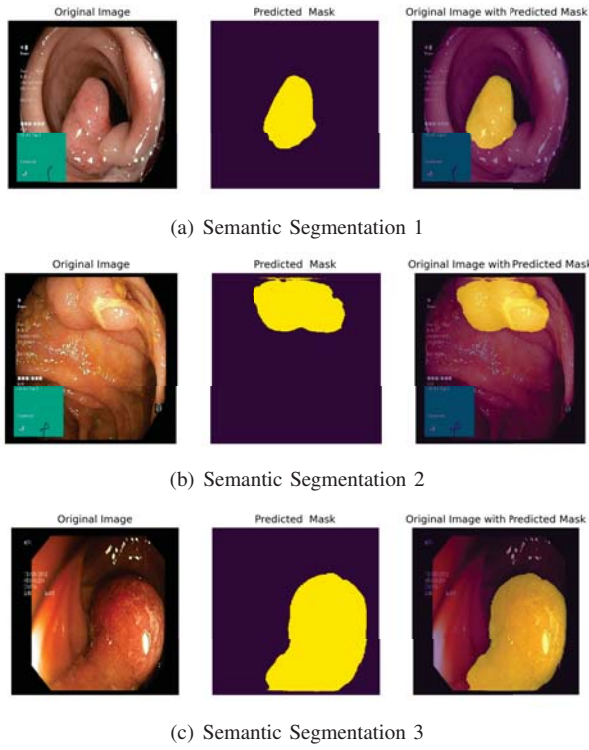


Fig. 6. Predicted mask from semantic segmentation on the test dataset.

masks are generated and saved in a folder. Some sample masks along with original images can be seen in Fig. 6.

For optimizing our model we have used optimizer and callback functions which reduce unnecessary high learning rates and adjust the weights of the layers automatically. Adam was used as the optimizer and in callback functions, we used CSVLogger, ModelCheckpoint, and learning rate reducer which saved computational cost. Due to these utility functions, we could achieve stability as well as a high value of accuracy early on in training. From the curves, it can be seen that the proposed model found the global minima after going through 30 epochs, and the model stabilized after around 45 epochs. The training accuracy of our model was 99.24% and validation accuracy was 94.83% where training and validation losses were recorded to be 1.75% and 21.42%, respectively. Training accuracy was slightly better than the validation accuracy as the images used for validation were not exposed to the model during training and this variation is within permissible limits.

D. IoU

The IoU is the most important parameter for any segmentation job. It describes how much area of any image is correctly binarized with reference to the available ground truth or reference mask as shown in (1) where P is the area of the predicted portion and G is the area of the ground truth portion.

$$\text{IoU} = \frac{\text{Area of Intersection}}{\text{Area of Union}} = \frac{|P \cap G|}{|P \cup G|} \quad (1)$$

In the segmentation task, accuracy may be misleading sometimes because the background of any medical image constitutes the major portion of the image compared to the main region of interest. Detecting this large background portion of the image constitutes the high accuracy of any segmentation model. That's why we calculate IoU in this case. For our model, we achieved an IoU score of 97.23% for training and 82.21% for validation. This IoU has surpassed the traditional machine learning model as well as the other state-of-the-art models developed by researchers having comparatively simplified model structures. We have plotted IoU against epochs shown in Fig. 7, which demonstrates that our model has a stabilized IoU and a high one also. The gap in training and validation IoU is acceptable in deep learning models as validation images were not shown to the model during training. A comparison of these metrics among the state-of-the-art models and our proposed hybrid model is shown in Table I.

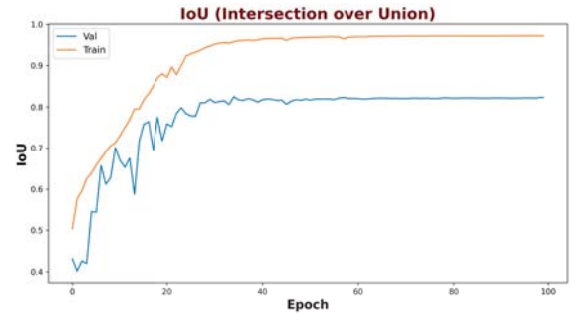


Fig. 7. IoU curve of the model.

TABLE I
METRICS COMPARISON AMONG DIFFERENT MODELS

Models	Evaluation Metrics		
	Accuracy	IoU	Dataset
U-Net [7]	85.67%	61.59%	Kvasir-SEG
PraNet [8]	88.23%	67.83%	Kvasir-SEG
ResUNet++ [9]	92.09%	79.27%	Kvasir-SEG
HarDNet [14]	89.64%	78.45%	Kvasir-SEG
Proposed Model	94.83%	82.21%	Kvasir-SEG, Kvasir V2

E. Watershed Mask

Finally, we generated the Watershed masks for all the binary masks of polyp class data of the Kvasir V2 dataset. We did some morphological operations on the binary masks such as the opening, closing, and distance transforms. These operations were necessary as part of the image preprocessing in this second stage of our proposed model. This provided us with meaningful information from each Watershed mask. We also adjusted the variable parameters of the Watershed algorithm to find out optimum values which define the boundaries of each

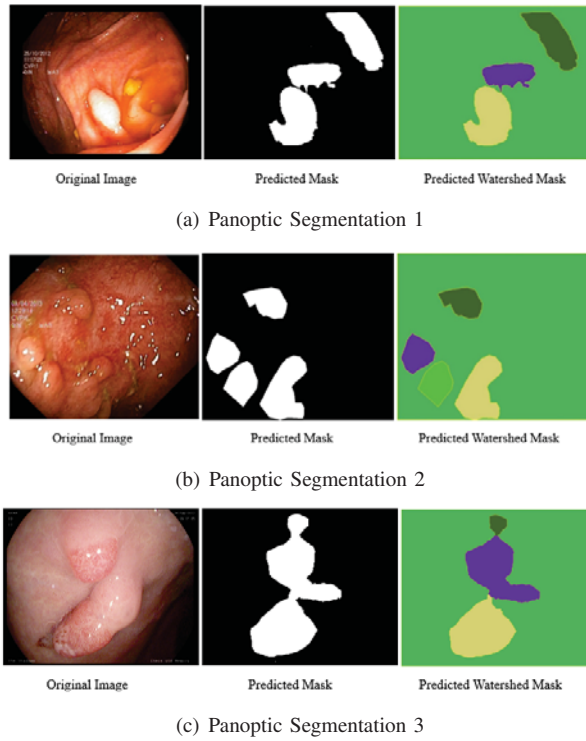


Fig. 8. Panoptic segmentation including semantic segmentation and Watershed mask.

object precisely along with assigning different pixel values for different objects even if they are of the same class. We cross-verified the masks with the original images as well as the binary masks which can be seen in Fig. 8. These results show how well our model performs to differentiate the polyps along with their sizes and locations. This helps greatly to decide the course of treatment for any patient.

V. CONCLUSION

This work presents the results of a hybrid deep learning-based model combining U-Net and Watershed algorithms to provide a complete solution to the panoptic segmentation of colorectal polyp images. After comparing its performance with other state-of-the-art models our model demonstrates superior performance, achieving a validation accuracy of 94.83% and a validation IoU score of 82.21%. The Watershed masks generated by our model are also good enough to segment different polyps with precise boundaries. Moreover, our proposed model is a simplified one and this has been optimized keeping

metrics and computational setup in mind. This model can be implemented in a moderate computational setup which reduces the computational cost as well as time. In the future, we plan to develop a more robust deep-learning model that can increase the metric scores for this panoptic segmentation and generate real-time binary as well as Watershed masks from the Hyper Kvasir video data.

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